ABTIN SHAHIDI

Observational Astrophysicist · Statistical Data Scientist

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Observational astrophysicist and statistical data scientist (Ph.D. Physics & Astronomy) with several years of industry experience building large-scale Bayesian and machine-learning systems. I develop probabilistic models for noisy, censored, and selection-biased data — from galaxy evolution in deep multiwavelength surveys to behavioral and demographic modeling across 120 million U.S. households.

RESEARCH INTERESTS

Observational extragalactic astrophysics · galaxy evolution, quenching, and the star-forming sequence · Bayesian hierarchical modeling · selection effects and censored data · machine learning and dimensionality reduction for astronomy · point processes and causal inference.

EXPERIENCE

Independent Researcher — 2026 – Present

Riverside, CA

- Independent research program in observational extragalactic astrophysics and applied probabilistic modeling, extending Bayesian hierarchical models of the galaxy star-forming sequence and the mass- and environment-dependence of quenching across cosmic time.

Postdoctoral Scholar — 2025 – 2026

University of California, Riverside

- Research in observational extragalactic astrophysics within the UCR Astronomy group, focused on statistical modeling of galaxy populations from multiwavelength photometric catalogs.

Senior Data Scientist — 2021 – 2024

VideoAmp

- **Demographic imputation across 120M U.S. households:** designed and implemented a Gibbs sampler for Bayesian non-parametric density estimation to impute missing values in nested household structures, scaling to ∼10M records and ∼1M parameters using *Hierarchical Dirichlet Process* (*HDP*) *mixture model* framework.
- **Identity-graph construction:** built a probabilistic framework modeling the time-evolution of interconnected bipartite identity graphs (∼100M nodes, ∼1B edges), integrating demographic data from multiple sources with an emphasis on geographic consistency.
- **Behavioral modeling:** developed hierarchical Bayesian non-homogeneous Poisson point-process models with autoregression, covariates and random effects to characterize purchase and TV-viewership behavior across 100,000+ time series.
- **Survey-based lift methodology:** engineered a selection-corrected projection method on purchase-panel data to estimate advertising effectiveness, accounting for identity-graph matchability and observability.

Graduate Research Assistant — 2017 – 2021

University of California, Riverside

- **Selection of massive evolved galaxies at $3 \leq z \leq 4.5$** across the CANDELS and COSMOS fields: Applied selection methods and quantified per-candidate confidence, completeness, and error propagation by combining observed colors, rest-frame UVJ colors, and SED-fitting–based approaches for CANDELS galaxies. Also led a large-scale statistical learning analysis of 1M galaxies using multiple machine learning techniques, with Santa Cruz semi-analytic model simulated catalogs serving as ground truth for training and validation.
- Developed hierarchical Bayesian models of the star-forming sequence under noisy and censored measurements, jointly inferring the SFR–mass relation, the stellar-mass function, and the mass/environment dependence of the quiescent fraction.

TEACHING

Teaching Assistant — 2016 – 2021

University of California, Riverside

- Introductory Physics & Laboratory, Interstellar Astrophysics, and Foundation of Applied Machine Learning (graduate course).
- Course materials and notebooks for Foundation of Applied Machine Learning are publicly available at github.com/abtinshahidi/Foundation_applied_machine_learning.

EDUCATION

Ph.D. in Physics & Astronomy — 2017 – 2021

University of California, Riverside

- Thesis: *Statistical Study of the Galaxy Populations at High Redshift from Multiwavelength Catalogs.*

M.Sc. in Physics & Astronomy — 2016 – 2017

University of California, Riverside

B.Sc. in Physics — 2011 – 2015

Sharif University of Technology, Tehran, Iran

PUBLICATIONS

Peer-reviewed; author name in **bold**.

1. Pacifici, C., Iyer, K. G., Mobasher, B., ... **Shahidi, A.**, et al. (2023). "The Art of Measuring Physical Parameters in Galaxies: A Critical Assessment of Spectral Energy Distribution Fitting Techniques." *The Astrophysical Journal*, 944(2), 141.
2. Hemmati, S., Huff, E., Nayyeri, H., Ferté, A., Melchior, P., Mobasher, B., Rhodes, J., **Shahidi, A.**, & Teplitz, H. (2022). "Deblending Galaxies with Generative Adversarial Networks." *The Astrophysical Journal*, 941(2), 141.
3. **Shahidi, A.**, Mobasher, B., Nayyeri, H., et al. (2020). "Selection of Massive Evolved Galaxies at $3 \leq z \leq 4.5$ in the

CANDELS Fields." *The Astrophysical Journal*, 897(1), 44. First author

4. Hemmati, S., Mobasher, B., Nayyeri, H., **Shahidi, A.**, et al. (2020). "Bridging between the Integrated and Resolved Main Sequence of Star Formation." *The Astrophysical Journal Letters*, 896(1), L17.
5. Darvish, B., ... **Shahidi, A.**, et al. (2020). "Spectroscopic Confirmation of a Coma Cluster Progenitor at $z \sim 2.2$." *The Astrophysical Journal*, 892(1), 8.
6. Chartab, N., ... **Shahidi, A.**, et al. (2020). "Large-scale Structures in the CANDELS Fields: The Role of the Environment in Star Formation Activity." *The Astrophysical Journal*, 890(1), 7.
7. Hemmati, S., Capak, P., ... **Shahidi, A.** (2019). "Bringing Manifold Learning and Dimensionality Reduction to SED Fitters." *The Astrophysical Journal Letters*, 881(1), L14.

HONORS & AWARDS

- **NASA MIRO FIELDS Fellowship** — Fellowships and Internships in Extremely Large Datasets (2019–2020).
- **Dean's Distinguished Fellowship**, University of California, Riverside (2016).
- **Silver Medal**, Iranian National Astronomy & Astrophysics Olympiad (2010).

TECHNICAL SKILLS

Programming	Python, R, Julia, SQL, Stan, MATLAB, Mathematica
Data & Systems	Spark / PySpark, Snowflake, Databricks, GNU/Linux, Git, LaTeX
ML & Stats Libraries	NumPy, Pandas, SciPy, scikit-learn, PyMC, PyStan, TensorFlow, scikit-tda, giotto-tda, Prophet
Methods	Bayesian hierarchical modeling, MCMC (Hamiltonian Monte Carlo, Gibbs sampling), point processes, statistical learning, manifold learning & dimensionality reduction, topological data analysis, SED fitting, causal inference, selection effects & censoring corrections